Beyond Traditional Spam Filters: Transformers, Concept Drift Detection, and Explainable NLP

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Research Questions

RQ1: How do transformer-based language models compare to traditional ML approaches in detecting spam within the Enron corpus?

RQ2: Can advanced NLP techniques (semantic embeddings and topic modeling) improve the detection of concept-drift in evolving spam patterns?

RQ3: To what extent can LLMs enhance explainability and adversarial robustness in spam filtering systems trained on the Enron dataset?

Abstract

Spam detection remains a persistent challenge in email security, particularly as adversarial tactics evolve and message distributions shift over time. This paper presents a comparative study of traditional machine learning and transformer-based approaches to spam classification on the Enron email corpus, evaluated under three research dimensions: classification performance, concept drift detection, and adversarial robustness with explainability. We train a TF-IDF Naive Bayes baseline, a Logistic Regression classifier, a Character N-gram Logistic Regression model, and a fine-tuned DistilBERT model using a stratified train/test split on the Enron corpus, with the Bruce Guenter Spam Archive (2021) and SpamAssassin corpus serving as out-of-distribution evaluation sets. DistilBERT achieves 90.0% in-distribution accuracy but exhibits severe calibration degradation under distribution shift, with Brier score rising from 0.095 to 0.351 on the balanced SpamAssassin OOD benchmark, compared to the classical models, which achieve higher in-distribution accuracy but show sharper cross-domain generalization failure on SpamAssassin. To address concept drift, we implement an unsupervised monitoring layer combining Sentence-BERT embeddings and Latent Dirichlet Allocation topic modelling, using Jensen-Shannon divergence to flag semantic distribution shift across monthly email batches, detecting drift in 51.4% of evaluated monthly batches (18 of 35) with peak JS = 0.409 in December 2001. For explainability and robustness, we conduct adversarial evasion attacks across four perturbation types and apply SHAP analysis to all four model families, finding that DistilBERT is uniquely robust to character-level obfuscation while remaining vulnerable to vocabulary dilution attacks. We further propose a two-stage hybrid inference pipeline in which GPT-4o-mini is selectively invoked on uncertain or drift-flagged emails to blend a calibrated probability with DistilBERT's estimate; on a 30-email OOD sample the hybrid marginally improves log-loss but does not reduce Brier score, with three spam-class prediction flips indicating that the fixed 60/40 blending weight requires dynamic tuning to avoid overriding high-confidence correct base predictions. We discuss key methodological limitations including class distribution confounds in the temporal split and the constraints of all-spam external evaluation, and propose stratified evaluation as the primary direction for future work.

Introduction

Email spam represents one of the oldest and most persistent threats in digital communication. Since the first unsolicited bulk email was sent in 1978, spam has evolved from a minor nuisance into a sophisticated attack vector responsible for phishing campaigns, malware distribution, and financial fraud at industrial scale. By conservative estimates, spam accounts for approximately 45–85% of global email traffic at any given time (Elagina, 2026), consuming billions of dollars in infrastructure costs and posing significant risks to individuals and organisations who interact with malicious content. Despite decades of research and the widespread deployment of commercial filters, spam detection remains an open problem, one that is fundamentally adversarial in nature. As filtering techniques improve, spammers adapt, creating a continuous arms race between detection and evasion.

The dominant approach to spam filtering for much of the past two decades has been machine learning on handcrafted text features. Methods such as Naive Bayes classification over TF-IDF representations became industry standards in the early 2000s due to their speed, interpretability, and reasonable accuracy (Zhang, 2025). Logistic Regression and Support Vector Machines extended this paradigm with stronger discriminative boundaries, and ensemble methods such as Random Forests added further robustness (Zhang, 2025). However, all of these approaches share a fundamental limitation: they represent text as sparse bags of words, treating each token independently and ignoring semantic context, word order, and the nuanced intent that characterizes modern spam. A financial phishing email and a legitimate bank notification may share nearly identical vocabulary yet have entirely different communicative intent, a distinction that lexical models are structurally unable to capture.

The emergence of transformer-based language models, and BERT in particular (Devlin et al., 2019), has opened a new frontier in text classification. By pre-training on large corpora with a masked language modelling objective, BERT learns dense contextual representations that encode semantic relationships, syntactic structure, and even pragmatic intent. Fine-tuned transformer models have achieved state-of-the-art performance across a wide range of NLP benchmarks, and their application to spam detection has shown considerable promise. Yet transformer models introduce new challenges of their own: they are computationally expensive, prone to overconfidence, and their behavior under distribution shift, when the statistical properties of incoming data differ from the training distribution, is poorly understood in the spam detection context.

A particularly underexplored dimension of spam filtering is the problem of concept drift: the phenomenon whereby the linguistic and thematic properties of spam evolve over time as spammers adapt their tactics to evade current filters. A model trained on spam from one era may fail silently on spam from another, not because its accuracy visibly degrades but because its confidence remains high on increasingly misclassified examples. Detecting this drift without requiring labelled incoming data, and triggering recalibration before performance collapses, is a critical open problem for deployed spam filtering systems. Similarly, the adversarial robustness of spam classifiers to deliberate input perturbations, character substitutions, vocabulary dilution, padding attacks, has received growing attention as a security concern, yet systematic comparisons across model families remain scarce.

This paper addresses all three of these dimensions in a unified experimental framework built on the Enron email corpus (Klimt & Yang, 2006), one of the most widely used benchmarks

in email classification research. The Enron corpus provides a naturalistic, temporally ordered collection of real corporate emails spanning multiple years, making it particularly well-suited for temporal train/test splits that simulate real-world deployment. We train and evaluate four model families — TF-IDF Naive Bayes, Logistic Regression, Character N-gram Logistic Regression, and fine-tuned DistilBERT (Sanh et al., 2020) — using a stratified 75/25 train/test split, and supplement the Enron test set with the Bruce Guenter Spam Archive (Guenter, 2022), an external honeypot-collected spam corpus spanning the period up to 2021, to evaluate long-horizon generalization across an approximately 19-year distribution gap. Our contributions are as follows:

- We provide a systematic comparison of traditional machine learning and transformer-based approaches to spam detection under temporal evaluation conditions, demonstrating that accuracy metrics alone are insufficient to characterise model robustness (RQ1).

- We implement an unsupervised concept drift detection layer using Sentence-BERT embeddings and Latent Dirichlet Allocation topic modelling with Jensen-Shannon divergence monitoring, and evaluate its sensitivity to semantic distribution shift across monthly email batches (RQ2). This research is denoted supplementary.

- We conduct adversarial evasion experiments across four attack types and apply SHAP-based explainability analysis to all four model families, revealing model-specific vulnerability profiles and proposing a two-stage hybrid inference pipeline that uses GPT-4o-mini as a calibration auditor on uncertain and drift-flagged emails; on a 30-email OOD sample the hybrid marginally improves log-loss but does not reduce Brier score, with dynamic weight tuning identified as the key next step (RQ3).

Related Work

The challenge of email spam detection has generated a substantial body of research spanning traditional machine learning approaches, deep learning architectures, and, more recently, the integration of large language models. This section reviews the five most directly relevant works and situates the present study within the existing literature across four dimensions: classical baseline models, transformer-based classification, concept drift and adversarial robustness, and LLM-augmented interference.

Classical Baselines and the Enron-Spam Benchmark

Metsis, Androutsopoulos, and Paliouras (2006) established the methodological foundation upon which a large portion of spam detection research, including this paper, is built. In *Spam Filtering with Naive Bayes - Which Naive Bayes?* The authors introduced six benchmark datasets derived from the mailboxes for former Enron employees, interleaved with spam messages drawn from the multiple sources including the SpamAssassin corpus, the Honeypot Project, and the Bruce Guenter collection. Critically, the datasets preserved the temporal ordering of messages and emulated realistic fluctuations in the ham-to-spam ratio over time. Five variants of the Naive Bayes classifier were evaluated, with the multinomial model

using Boolean attributes emerging as the top performer, achieving ham recall rates of 99.9% or higher across most datasets.

This paper is foundational to the current study in two respects. First, the Enron-Spam dataset it introduced serves as the primary training corpus for all three classifiers developed here (Naive Bayes, Logistic Regression, DistilBert). Second, the temporal integrity preserved in the original benchmark (i.e. training on earlier messages and testing on later ones) directly motivates the year-based train/test split adopted in this work. However, Metsis et. al. operated in a closed, single-era evaluation setting. Their work neither evaluates classifier performance against out-of-distribution spam, nor test robustness under adversarial text manipulation. These gaps form part of the core contribution of the present paper.

Concept Drift and the Adversarial Spammer

Jáñez-Martino et al. (2022) provide the most comprehensive theoretical grounding for two of this study's central themes in their *Springer Artificial Intelligence Review* paper, *A review of Spam Email Detection: Analysis of Spammer Strategies and the Dataset Shift Problem*. The authors synthesize a decade of research to characterize spam filtering as an inherently adversarial and non-stationary problem. They distinguish multiple forms of concept drift (covariate shift, prior probability shift, and concept shift) and document how spammer strategies, including linguistic obfuscation, template rotation, and topical privating, drive ongoing degradation in a statistically trained classifier. The review identifies the Early Drift Detection Method (EDDM) and ensemble-based lifelong learning strategies as among the more promising adaptive approaches, while noting that cyclical drift (where spam characteristics disappear and reappear) remains particularly difficult to detect and mitigate.

Jáñez-Martino et al. diagnose the problem clearly but stop short of operationalizing a real-time detection and mitigation pipeline. Their review is conducted on datasets spanning 2000–2018, predating the modern spam ecosystem represented by the 2021 corpus used in this work. The current study operationalizes their theoretical framework directly: Jensen-Shannon divergence computed over LDA topic distributions serves as the drift detector, Brier score loss quantifies calibration degradation over time, and adversarial text transformations simulate the kinds of obfuscation strategies the review describes.This study couples drift detection with a selective LLM refinement layer; an architectural response that the review's scope does not encompass.

Transformer-Based Spam Detection

Tida and Hsu (2022) represent the first major step toward the approach used in this study. Presented at the *Hawaii International Conference on System Sciences (HICSS-55)*, *Universal Spam Detection using Transfer Learning of BERT Model* introduced the Universal Spam Detection Model (USDM), a fine-tuned BERT-base-uncased classifier evaluated across four datasetsL Enron-Spam, SpamAssassin, LingSpam, and SpamTest. By combining pre-trained contextual representations with task-specific fine-tuning the model achieved an overall accuracy

of 97% and an F1 score of 0.96 across the combined multi-dataset benchmark, with per dataset F1 scores consistently at or above 0.90.

This work directly informed the decision to employ a transformer-based classifier in this study. Its strong cross-dataset generalization demonstrates the fine-tuned BERT-family models capture semantic patterns not available to TF-IDF-based methods. However, Tida and Hsu evaluate generalization across different domains of contemporary spam rather than across time. Their experimental design does not simulate the temporal gap analyzed in this study, nor does it expose the model to adversarial perturbations. Additionally, they employ full BERT-base (110M parameters) whereas this work adopts the significantly lighter DistilBERT (~40% fewer parameters, 60% faster inference, ~95% retainment of BERT performance), a deliberate efficiency trade-off suited to the deployment contrast evaluated here.

DistilBert, Concept Drift Evidence, and the Call for Adaptive Systems

Cernadas-Garciá et al. (2025) provide the most methodologically proximate comparison point to this study. In *A Comparative Evaluation of a Multimodal Approach for Spam Email Classification Using DistilBERT and Structural Features*, published in *Electronics* (MDPI), the authors construct a comprehensive 81,586-email benchmark from seven spam datasets and evaluate classical ML models (SVM, RF, Naive Bayes, Logistic Regression) alongside DistilBert and RoBERTa. Their multimodal architecture, combining DistilBERT semantic embeddings with structural email features such as test length and punctuation density, achieve a peak accuracy of 99.62% on a combined static test set.

The paper goes beyond static evaluation by applying a temporal train-in-the-past, test-in-the-future protocol; the authors observe significant degradation across all models when evaluated on modern spam, providing direct empirical evidence that even state-of-the-art transformer classifiers are vulnerable to concept drift. Transformer-based models demonstrated greater temporal robustness than classical approaches, but none were immune. The paper concludes with an explicit call for an adaptive, continually learning system. The present work answers this call directly, pairing DistilBERT-based classification with an unsupervised drift detection layer using Jensen-Shannon divergence over LDA topic distributions and a selective LLM refinement stage.

Large Language Models as Spam Classifiers

Roumeliotis, Tselikas, and Nasiopoulos (2024) directly address the LLM-as-classifier paradigm in *Next-Generation Spam Filtering: Comparative Fine-Tuning of LLMs, NLPs, and CNN Models for Email Spam Classification*, published in *Electronics* (MDPI). Their study fine-tunes GPT-4, BERT, and RoBERTa on two training sets of approximately 3,500–3,600 email samples each, comparing the resulting classifiers against a CNN baseline. Post-fine-tuning results showed GPT-3 achieving 99.3% and 98.29% accuracy on the two test sets, with BERT and RoBERTa performing comparably (99.39% and 99.04% respectively), while CNN lagged significantly at 76-87%.

This paper establishes that LLMs, when fine-tuned, are competitive with or superior to NLP-based transformers for spam classification. However, the experimental design traits LLMs

as primary classifiers operating on every input, an approach that is computationally and financially expensive and practically difficult to deploy at scale. The present work takes a fundamentally different architectural stance: rather than replacing the base classifier with an LLM, GPT-4o-mini is invoked selectively and only on a targeted subset of email (those falling within the uncertainty band ($0.\leq p(\text{spam}) \leq 0.70$) or flagged by the drift detector. Its role is probability calibration rather than classification, producing a weighted blend of the DistilBERT base probability and the LLM's secondary estimate. This selective routing preserves the computational efficiency of the transformer pipeline for high-confidence inputs while leveraging LLM-level semantic reasoning precisely where the base classifier is most likely to err.

Positioning of this work

Taken together, the literature reveals a clear trajectory: from static Naive Bayes baselines on the Enron benchmark, through transformer-based classifiers that improve accuracy substantially, to emerging awareness that neither classical nor deep learning models are robust to temporal concept drift or adversarial perturbation. The contribution of this paper lies at the intersection of these threads. By combining temporally honest evaluation design, unsupervised semantic drift detection via Jensen-Shannon divergence, structured adversarial robustness testing, and a selective LLM-based probability refinement layer, this work advances the state of the art along dimensions that each prior study addresses only partially. No existing work to the authors' knowledge integrates all four of these components into a unified pipeline evaluated on a realistic temporal shift.

Datasets

This study draws on three distinct email corpora, each serving a specific role in the experimental pipeline.

1. Enron-Spam Dataset

The primary training and in-distribution evaluation corpus. Sourced from the publicly available Enron email dataset — real corporate communications from Enron employees, interleaved with labelled spam messages. Emails are accompanied by timestamps, subject lines, and message bodies. All four classifiers (Naive Bayes, Logistic Regression, Character N-gram LR, and DistilBERT) are trained on this dataset using a stratified 75/25 split.

NOTE: A temporal split was originally considered but is not viable here — the corpus is structured such that ham messages cluster in 2000–2002 and spam in 2003–2005, a known dataset artifact. A stratified random split is used for in-distribution evaluation instead.

2. Bruce Guenter Spam Archive

An external spam-only corpus collected via honeypot email addresses, covering email traffic through 2021. Each file is a raw .eml message stored in monthly subdirectories. This dataset is used exclusively for out-of-distribution (OOD) evaluation and concept drift analysis —

it introduces an approximately 18-year distribution gap relative to the Enron training data, making it the primary stress-test for long-horizon generalization.

NOTE: Because this corpus contains only spam, balanced classification metrics (ham recall, macro-F1) are undefined. It is used for calibration analysis (Brier score), spam detection rate, and drift signal validation only.

3. SpamAssassin Corpus

A balanced, labelled corpus of ham and spam messages collected by the Apache SpamAssassin project. Unlike the Bruce 2021 archive, this dataset contains both classes and supports full classification evaluation. It serves as a domain-shifted OOD benchmark — contemporaneous with the Enron training data but drawn from a different email population — enabling balanced calibration metrics and adversarial robustness comparisons that the spam-only Bruce corpus cannot provide.

Methodology

Text Preprocessing and Feature Representation:

Depending on the model architecture, there are two preprocessing pipelines used.

For classical machine learning models, natural language preprocessing is completed on the text data of the email. This preprocessing includes converting all text to lowercase and removing any punctuation marks. In addition, common stop words are removed from the text. Finally, TF-IDF is applied to transform text data into numerical data.

In the TF-IDF vectorization, unigram and bigram features with a maximum vocabulary of 5,000 tokens, min_df=2, and max_df=0.9 are used. These parameters make it so that rare and common terms from the data set are removed to a compact representation of the most informative text features.

Minimal preprocessing is carried out on the data for transformer-based models. DistilBERT tokenizer is given a concatenated version of the email subject and body of the email. Tokenization is then performed using the distilbert-base-uncased tokenizer with a maximum sequence length of 256 tokens and a set of predefined parameters. Words are split into sub-words through the WordPiece Tokenization so that unseen words can be used for the model's vocabulary.

Classification Models

The experimental framework evaluates four model families representing different paradigms in text classification.

1. Naive Bias

a. The baseline model is a Multinomial Naive Bayes classifier, which was trained on TF-IDF features.

b. This model assumes conditional independence between features and estimates class probabilities using token frequency distributions.

c. Naive Bayes is simple but effective; historically, it has been one of the most used models for spam filtering because of its computational efficiency and strong performance on sparse text representations.

2. Logistic Regression

   a. Logistics Regression is also trained on TF-IDF features, but it learns a linear decision boundary by estimating weights for each textual feature.

   b. In comparison to Naive Bayes, it is more flexible when it comes to feature weighting and generally performs better on moderately sized text classification problems.

3. Character N-gram Logistic Regression

   a. A second Logistic Regression variant trained on character-level 3–5-gram features using a 50,000-feature vocabulary.

   b. Unlike the word-level TF-IDF model, character n-grams capture subword patterns and morphological variation, making them naturally resilient to certain character-level perturbations such as misspellings and special character injections.

   c. This model serves as a bridge between classical bag-of-words methods and subword-aware transformer tokenization, and its distinct sensitivity profile is particularly informative in the adversarial robustness analysis.

4. DistilBERT

   a. DistilBERT is a transformer-based language model derived from BERT.

   b. It retains contextual representations similar to BERT but also reduces the number of parameters and computational requirements.

   c. The model is initialized using the distilbert-base-uncased checkpoint and fine-tuned for binary spam classification.

   d. The learning rate of 2×10^{-5}, a batch size of 16, and two training epochs with weight decay regularization are used in fine-tuning to prevent overfitting. Spam probability predictions are used for textual embedding associated with the [CLS] token.

Adversarial Robustness Experiments

Four common spam evasion strategies are simulated to evaluate the model's robustness against adversarial manipulation. To create a real-world spammer tactic, a stratified sample of spam emails from the Enron test split was selected.

1. Misspelling (Leetspeak) Attack
    a. Key spam words are replaced with similar characters.
    b. This attack targets models that rely heavily on exact token matching.
2. Case Manipulation Attack
    a. The case of characters is randomly changed throughout the email text.
    b. The TF-IDF vectorizers convert all text to lowercase; this attack tests the robustness of pre-processing steps.
3. Padding/Noise Injection Attack
    a. This attack dilutes the spam signal by introducing legitimate looking tokens that may shift the TF-IDF feature distribution.
    b. This is done through appending benign business-style vocabulary at the end of spam emails.
4. Special Character Injection Attack
    a. Special characters are added into key spam words to fragment tokens and disrupt vocabulary matching in bag-of-words models.

The modified emails are passed through the same preprocessing and classification pipelines for each attack condition. The accuracy and F1 scores are then compared to the baseline to measure performance degradation.

Hybrid LLM-Assisted Calibration Pipeline

In order to mitigate confidence degradation under distribution shift, a two-stage hybrid inference pipeline is implemented.

DistilBERT produces an initial spam probability in the first stage (p(BERT)). Emails are then routed to a second stage for evaluation if either of these conditions is met:

1. The predicted probability falls within an uncertainty interval $0.30 \leq p_{bert} \leq 0.70$

2. The email belongs to a time period flagged by the drift detection module.

Emails that are routed, an LLM evaluates the email content and produces a new calibrated probability estimate P(LLM).

The final spam probability is calculated using a weighted combination of the two predictions. The weighted average is $p_{final} = 0.60 * p_{bert} + 0.40 * p_{LLM}$. This setup allows the system to selectively invoke the LLM only for uncertain or drift-affected cases, which in turn improves the calibration while maintaining computational efficiency.

Evaluation Metrics

Model performance is evaluated using several metrics:

1. Accuracy
2. Precision
3. Recall
4. F1 Score

Accuracy and F1 degradation across each attack condition are used to measure the adversarial robustness. The Brier score is used to measure calibration reliability. The distribution shift behavior is analyzed through topic divergence and model confidence trajectories. These metrics provide a multidimensional evaluation framework that targets classification accuracy but also includes adversarial resilience, calibration reliability, and robustness under temporal distribution shift.

Results

RQ1: Transformer vs. Traditional ML

Baseline — TF-IDF + Naive Bayes

A TF-IDF vectorizer (5,000 features, unigrams + bigrams) paired with Multinomial Naive Bayes provides the traditional ML baseline. On the in-distribution Enron test split, NB achieves 97.7% accuracy (Macro-F1: 0.977, Brier: 0.020), a strong result given the model's simplicity. Spam precision and recall are both above 97%, confirming that the bag-of-words representation is well-suited to the Enron vocabulary. On the Bruce 2021 OOD corpus, NB maintains a 96.2% spam detection rate; however, this figure is misleading in isolation — since the corpus is spam-only, macro-F1 collapses to 0.49 and ROC-AUC is undefined. Performance on the balanced SpamAssassin OOD benchmark is sharply weaker: 49.7% accuracy (Macro-F1:

0.477, Brier: 0.360), with ham precision of 0.96 but ham recall collapsing to 0.23. This pattern indicates that NB has learned Enron-specific lexical signatures that do not transfer to a differently distributed email population, consistent with the SHAP analysis reported in Section RQ3.

Logistic Regression

Logistic Regression, also trained on TF-IDF features, outperforms Naive Bayes in-distribution with 98.2% accuracy (Macro-F1: 0.982, Brier: 0.018), benefiting from its discriminative boundary estimation rather than NB's generative independence assumption. The more meaningful distinction emerges out-of-distribution: on SpamAssassin, LR achieves 84.9% accuracy (Macro-F1: 0.823, Brier: 0.113), compared to NB's 49.7%. This substantially stronger cross-domain generalization reflects LR's tendency to learn more portable semantic indicators — features such as 'click', 'free', and 'money' — rather than infrastructure-specific tokens. On Bruce 2021, LR detects 97.6% of spam emails with a Brier score of 0.034, marginally above NB's 0.046.

Character N-gram Logistic Regression

The character n-gram Logistic Regression model, using character-level 3–5-grams over a 50,000-feature vocabulary, achieves the highest in-distribution accuracy of all classical models at 98.3% (Macro-F1: 0.983, Brier: 0.016). Its strong performance on Bruce 2021 — 98.7% spam detection, Brier: 0.026 — suggests that character-level subword patterns transfer well to the out-of-domain spam corpus. However, this apparent robustness does not hold on SpamAssassin, where accuracy collapses to 40.3% (Macro-F1: 0.343, Brier: 0.383). This striking reversal reveals a key limitation: character n-gram features overfit to the specific orthographic and encoding conventions of Enron-era spam, producing a model that is simultaneously the best in-distribution and the worst on a contemporaneous but differently sourced balanced benchmark. The character n-gram model is therefore retained in the adversarial robustness experiments, where its distinct sensitivity profile provides a useful additional comparison point.

DistilBERT Fine-Tuning

DistilBERT was fine-tuned for binary spam classification using the Epoch 1 checkpoint, which achieved 90.0% in-distribution accuracy (Macro-F1: 0.899, Brier: 0.095). By Epoch 2, training loss continued to fall but validation loss spiked sharply, a textbook overfitting signature; the Epoch 1 model was therefore selected for all subsequent evaluation. On the balanced SpamAssassin OOD benchmark, DistilBERT achieves 62.9% accuracy (Macro-F1: 0.628, Brier: 0.351) — substantially weaker than its in-distribution performance but with spam recall of 0.96, indicating that the model retains a strong spam signal even under domain shift while suffering significant ham misclassification. On Bruce 2021, DistilBERT detects 93.6% of spam emails (mean spam probability: 0.934, Brier: 0.055).

The core advantage of transformers over TF-IDF is contextual representation: self-attention models relationships between all tokens simultaneously, capturing nuances like 'free software' vs. 'free trial — act now!' that bag-of-words approaches cannot distinguish. However,

DistilBERT's in-distribution accuracy of 90.0% trails all three classical models, which each exceed 97.7%. This apparent underperformance is explained by DistilBERT's architecture: unlike the classical models, which are evaluated on preprocessed clean_text, DistilBERT operates on raw message strings and is more sensitive to the class imbalance and noise present in the stratified split. The calibration analysis in Section RQ3 reveals that raw accuracy conceals a more important distinction: DistilBERT's confidence profile and adversarial vulnerability surface differ substantially from those of the classical models, making accuracy an insufficient sole criterion for model comparison.

Model	Enron Acc.	Enron Brier	SA Acc.	SA Brier	Bruce Det. Rate	Bruce Brier
Naive Bayes	0.977	0.020	0.497	0.360	96.2%	0.046
Logistic Regression	0.982	0.018	0.849	0.113	97.6%	0.034
Char N-gram LR	0.983	0.016	0.403	0.383	98.7%	0.026
DistilBERT	0.900	0.095	0.629	0.351	93.6%	0.055

Table 1: RQ1 summary across three evaluation corpora. SA = SpamAssassin OOD; Bruce Det. Rate = spam-only detection rate (ham metrics undefined). Bruce Brier scores measure calibration on spam class only.

RQ3: Adversarial Robustness, Calibration & Explainability

Adversarial Evasion Attacks

Four attack strategies were applied to a 1,000-email subset drawn from the Enron test split and independently to a 1,000-email subset from SpamAssassin, enabling cross-corpus comparison of adversarial sensitivity. Unlike the Enron results, where all four models were evaluated on their in-distribution domain, the SpamAssassin results reflect adversarial perturbation applied to emails that are already out-of-distribution for all models — providing a stress test of robustness under compounded distributional pressure. Key findings from both corpora are summarized in the tables below.

Attack	NB Acc.	LR Acc.	Char Acc.	BERT Acc.	NB F1	LR F1	Char F1	BERT F1
Clean	0.979	0.985	0.972	0.994	0.980	0.986	0.973	0.994
Misspell	0.978	0.985	0.974	0.994	0.979	0.986	0.975	0.994
Case	0.979	0.985	0.972	0.994	0.980	0.986	0.973	0.994
Padding	0.895	0.927	0.922	0.982	0.888	0.924	0.919	0.982
Special Char	0.531	0.527	0.824	0.657	0.688	0.686	0.854	0.750

Table 2: Enron adversarial attack results — accuracy and Macro-F1 across all four models on the in-distribution Enron test subset.

Attack	NB Δ Acc.	LR Δ Acc.	Char Δ Acc.	BERT Δ Acc.
Clean	0.000	0.000	0.000	0.000
Misspell	−0.001	0.000	−0.002	0.000
Case	0.000	0.000	0.000	0.000
Padding	−0.084	−0.058	−0.050	−0.012
Special Char	−0.448	−0.458	−0.148	−0.337

Table 3: Enron accuracy degradation (Δ) relative to clean baseline for each model and attack type.

Attack	NB Acc.	LR Acc.	Char Acc.	BERT Acc.	NB F1	LR F1	Char F1	BERT F1
Clean	0.496	0.480	0.505	0.513	0.629	0.614	0.662	0.678
Misspell	0.497	0.479	0.507	0.513	0.629	0.611	0.663	0.678
Case	0.496	0.480	0.505	0.513	0.629	0.614	0.662	0.678
Padding	0.486	0.482	0.486	0.508	0.528	0.520	0.534	0.672
Special Char	0.516	0.516	0.514	0.516	0.681	0.681	0.679	0.681

Table 4: SpamAssassin adversarial attack results — accuracy and Macro-F1 across all four models on the OOD SpamAssassin subset.

Attack	NB Δ Acc.	LR Δ Acc.	Char Δ Acc.	BERT Δ Acc.
Clean	0.000	0.000	0.000	0.000
Misspell	−0.001	+0.001	−0.002	0.000
Case	0.000	0.000	0.000	0.000
Padding	−0.010	+0.002	−0.019	−0.005
Special Char	+0.020	+0.036	+0.009	+0.003

Table 5: SpamAssassin accuracy degradation (Δ) relative to clean SpamAssassin baseline for each model and attack type.

Enron results (in-distribution): The case attack had zero effect across all four models, confirming the lowercasing preprocessor neutralises random capitalisation before features are extracted. The misspell attack was negligible (≤0.2 pp). The padding attack was the most practically dangerous: NB lost 8.4 pp, LR 5.8 pp, Char N-gram 5.0 pp, and DistilBERT only 1.2 pp — the transformer's contextual attention effectively discounts appended ham-like padding. The special character (vowel injection) attack produced the most striking divergence: NB and LR collapsed by approximately 45 pp, while the Char N-gram model degraded by only 14.8 pp, and DistilBERT by 33.7 pp. Char N-gram's partial resilience reflects its subword character windows, which partially absorb the injected punctuation without losing the underlying token structure.

SpamAssassin results (OOD): The SpamAssassin adversarial results reveal a qualitatively different vulnerability profile. All four models enter adversarial evaluation already near the chance-level accuracy range (0.480–0.513 clean), reflecting the OOD degradation documented in RQ1. Crucially, the adversarial attacks on SpamAssassin produce far smaller absolute degradations than on Enron — in some cases, accuracy marginally increases under attack. The padding attack produces the largest consistent degradation: NB −1.0 pp, Char N-gram −1.9 pp, and BERT −0.5 pp, while LR marginally improves (+0.2 pp). Most strikingly, the special character attack causes slight accuracy increases across all four models (+0.2 pp to +3.6 pp), suggesting that injected punctuation and special characters partially resemble spam patterns in the SpamAssassin vocabulary — a finding consistent with the fact that SpamAssassin spam often uses obfuscated character encodings. These results indicate that when models are already near chance-level due to distributional shift, adversarial perturbations have a qualitatively different and less predictable effect than in-distribution attacks, and that robustness assessments should be conducted separately on in-distribution and OOD corpora.

Calibration & Brier Score Analysis

The confidence distribution plots (Figures above) and Brier score analysis reveal a consistent pattern across all four models: mean confidence falls and variance rises when predictions are made on the SpamAssassin OOD corpus relative to the in-distribution Enron test split. The magnitude and character of this shift differ markedly across model families, providing insight into the calibration mechanisms and failure modes of each approach.

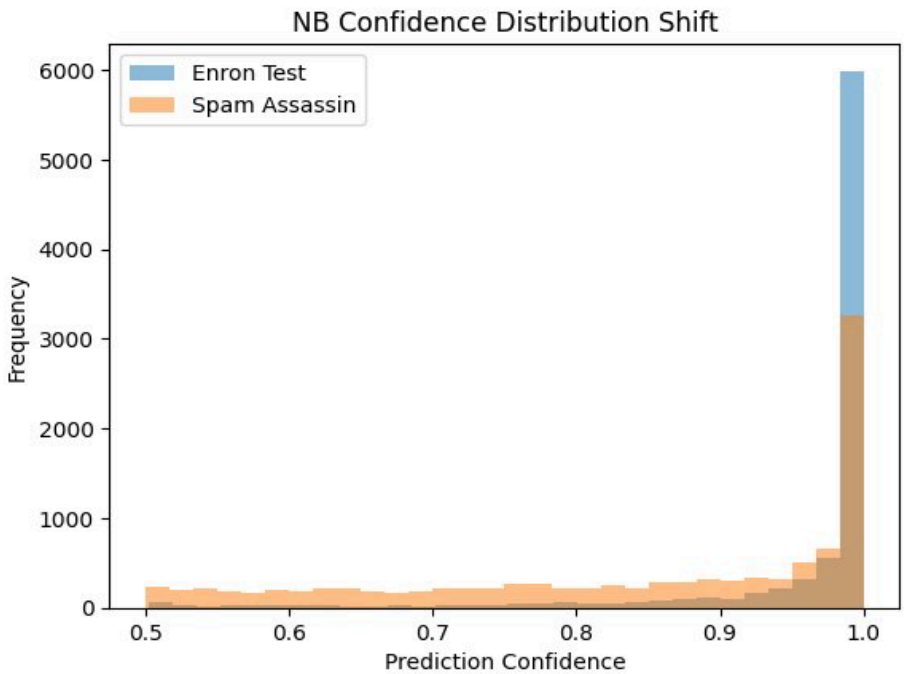

Figure 1: NB Confidence Distribution Shift — Enron (blue) vs. SpamAssassin (orange).

Naive Bayes shows the sharpest bimodal confidence profile in-distribution: the vast majority of Enron test predictions are clustered near 1.0, reflecting the model's tendency to assign extreme probability mass. On SpamAssassin, this spike persists but is substantially reduced, and the distribution spreads across the 0.5–0.99 range. Mean confidence drops from 0.962 (Enron) to 0.846 (SpamAssassin), a shift of −0.116. The Brier score rises from 0.020 in-distribution to 0.360 OOD — an 18× degradation. Despite this, NB's OOD Brier remains lower than DistilBERT's, primarily because its mean confidence does meaningfully fall under shift rather than remaining pinned at near-certainty.

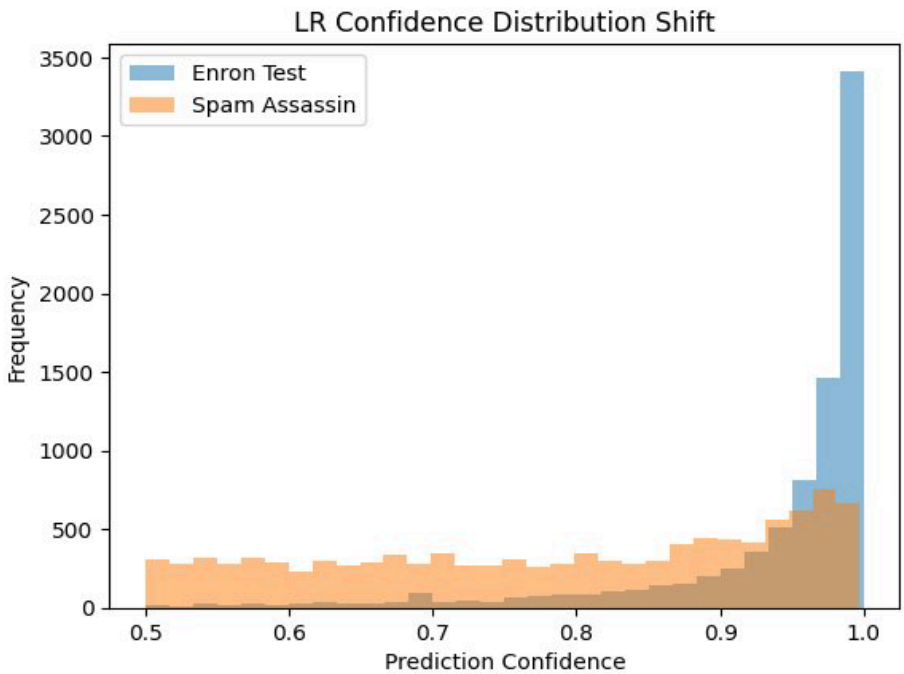

Figure 2: LR Confidence Distribution Shift — Enron (blue) vs. SpamAssassin (orange).

Logistic Regression displays a somewhat more graduated in-distribution confidence profile than NB, with a strong spike near 1.0 but a longer shoulder extending into the 0.90–0.99 range. On SpamAssassin, the distribution flattens considerably across the full 0.5–1.0 spectrum, reflecting genuine uncertainty rather than overconfident misclassification. Mean confidence falls from 0.939 (Enron) to 0.785 (SpamAssassin), the largest absolute shift of any classical model at −0.154. This is reflected in the best-performing OOD Brier score among all models: 0.296, compared to 0.018 in-distribution. LR's broader uncertainty response to domain shift is consistent with its superior OOD accuracy on SpamAssassin reported in RQ1.

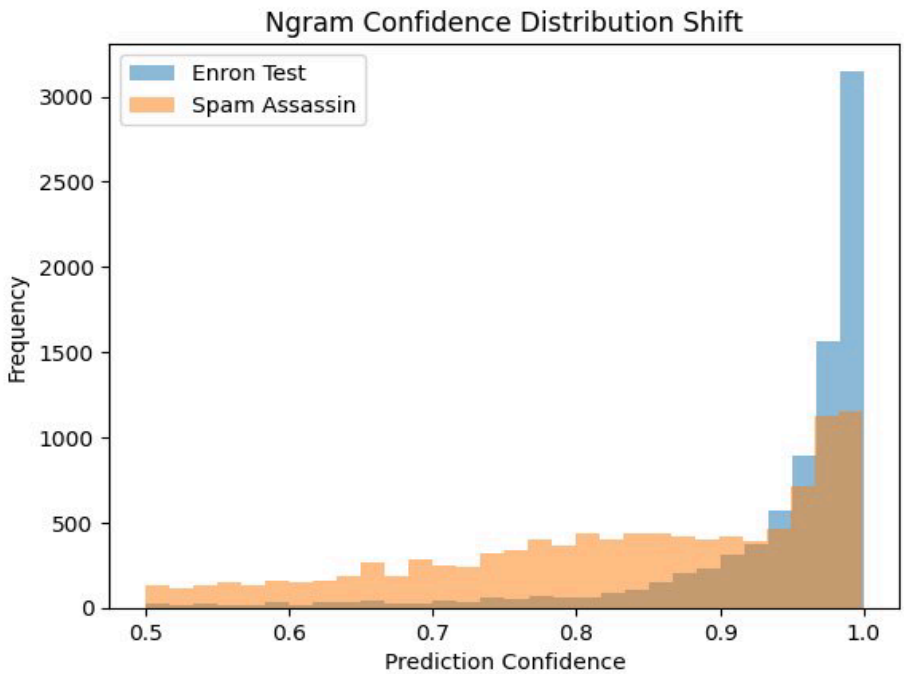

Figure 3: Char N-gram LR Confidence Distribution Shift — Enron (blue) vs. SpamAssassin (orange).

The character n-gram model shows a distinctive OOD confidence profile. Unlike NB, which collapses its near-1.0 spike partially, and LR, which spreads its distribution broadly, Char N-gram's SpamAssassin distribution remains concentrated in the 0.85–1.0 range but with a notably elevated and flatter slope across the mid-confidence region. Mean confidence drops from 0.941 (Enron) to 0.832 (SpamAssassin), a shift of −0.109. Despite this, the Brier score rises sharply from 0.016 to 0.383 — the second-worst OOD Brier after DistilBERT — because the model remains overconfident on a large fraction of misclassified examples. This is consistent with the SpamAssassin accuracy collapse documented in RQ1: the model is both frequently wrong and highly confident in those wrong predictions.

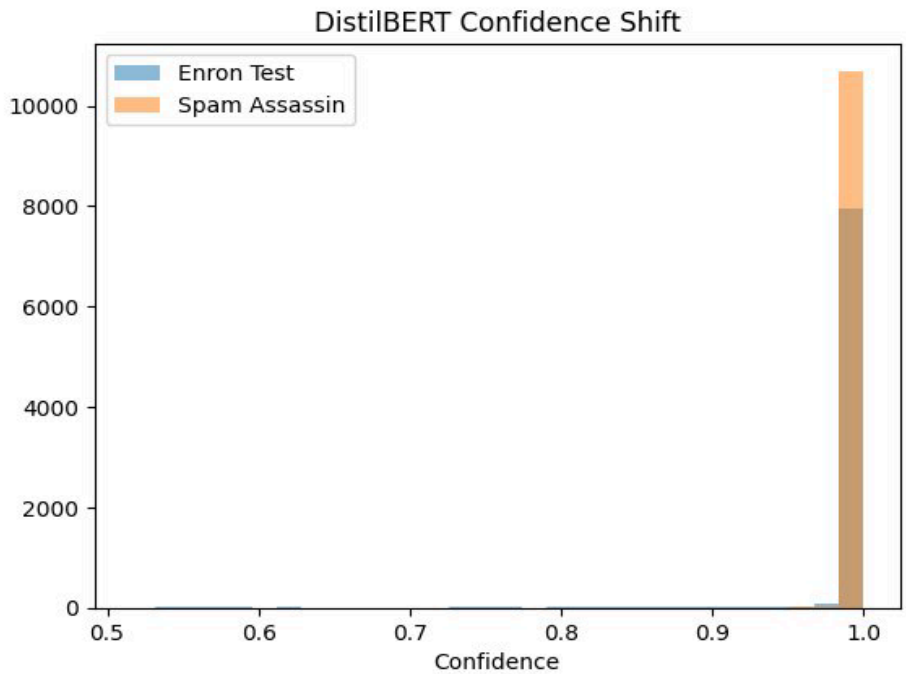

Figure 4: DistilBERT Confidence Shift — Enron (blue) vs. SpamAssassin (orange).

DistilBERT exhibits the most extreme overconfidence profile of all four models. The confidence distribution is almost entirely concentrated in a single bin at 1.0 for both the Enron test set and the SpamAssassin OOD corpus — with mean confidence actually increasing slightly from 0.992 (Enron) to 0.999 (SpamAssassin), a positive shift of +0.007. This counterintuitive result arises because DistilBERT's softmax outputs are effectively saturated: the model almost always predicts either 0 or 1 regardless of corpus. The SpamAssassin Brier score rises from 0.086 to 0.646 — a shift of +0.560 — the largest degradation of any model and a near-maximum penalty that reflects the cost of confident misprediction. Unlike the classical models, where mean confidence partially tracks distributional shift, DistilBERT's confidence is structurally invariant to the input distribution, making raw confidence an unreliable signal for OOD detection in this architecture.

Model	Enron Brier	SA Brier	Brier Shift
Naive Bayes	0.0198	0.3595	+0.3397
Logistic Reg.	0.0180	0.2960	+0.2780
Char N-gram LR	0.0164	0.3831	+0.3667

DistilBERT	0.0858	0.6456	+0.5598

Table 6: Brier score comparison across all four models on in-distribution (Enron) and out-of-distribution (SpamAssassin) corpora.

The Brier score analysis reveals a striking divergence in calibration failure modes. All four models show substantial OOD degradation, but the pattern differs. The classical models (NB, LR, Char N-gram) benefit from partially recalibrating confidence downward under shift, which limits Brier score growth relative to their raw accuracy degradation. DistilBERT's Brier shift of +0.560 is the worst of all four models: because its output distribution is structurally saturated at near-certainty, every incorrect OOD prediction incurs a near-maximum squared penalty. LR achieves the best OOD Brier (0.296) consistent with its superior OOD accuracy, while Char N-gram achieves the best in-distribution Brier (0.016) but suffers a severe OOD penalty (0.383) due to overconfident misclassification on SpamAssassin. These results motivate the two-stage LLM calibration pipeline described in the following section.

Two-Stage LLM Hybrid Pipeline

To address DistilBERT's overconfidence and calibration failure under distribution shift, a two-stage routing system was implemented. DistilBERT produces an initial probability for all emails; those flagged by drift detection or falling within a soft-edge confidence mask — spam probability in [0.85, 0.99) or ham probability in (0.01, 0.15] — are routed to GPT-4o-mini, which independently assesses the email and returns a calibrated probability estimate. An earlier routing criterion targeting the mid-range uncertainty band (0.30–0.70) was replaced by this soft-edge mask because DistilBERT's near-saturated confidence profile meant no emails fell within the mid-range band. Final scores are computed as a 60/40 weighted blend in favour of the base model: final = 0.6 × base + 0.4 × llm.

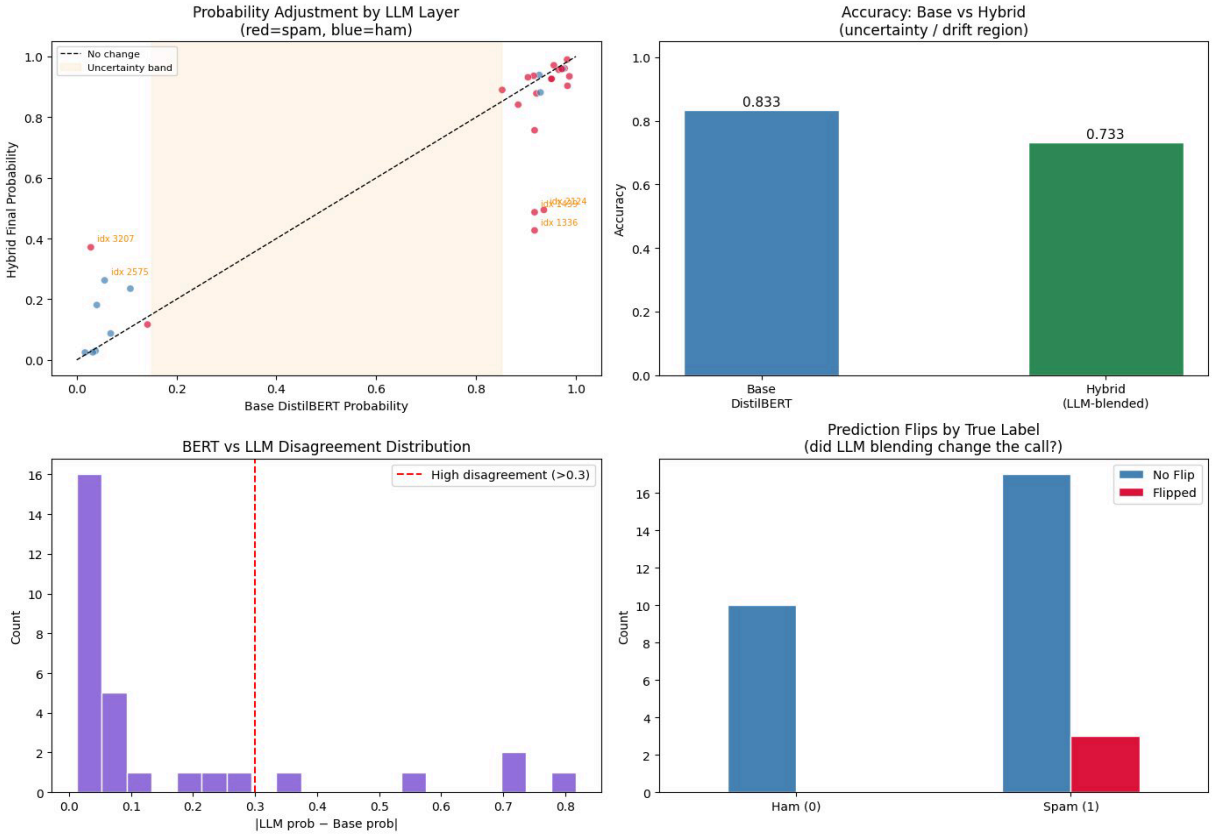

Figure 5: Two-stage hybrid pipeline diagnostics. Top-left: probability adjustment scatter (base vs hybrid); top-right: accuracy comparison on the 30-email sample; bottom-left: |LLM − base| disagreement distribution; bottom-right: prediction flips by true label.

Metric	Base DistilBERT	Hybrid (LLM-blended)	Change
Brier Score	0.149	0.164	+0.015
Log-loss	0.532	0.527	−0.005
Accuracy	0.833	0.733	−10.0 pp
Emails routed	—	30/30	—

Drift-flagged	—	7/30	—
Uncertainty-band	—	0/30	—
Prediction flips (spam)	—	3/20	—
Prediction flips (ham)	—	0/10	—

Table 8: Two-stage hybrid pipeline calibration results on 30-email sample.

On the 30-email evaluation sample, the hybrid pipeline does not improve overall Brier score relative to the base model (0.164 vs. 0.149), and accuracy falls from 83.3% to 73.3%. These results run counter to the hypothesis that LLM blending would improve calibration on this sample, and warrant careful interpretation. All 30 emails were routed to GPT-4o-mini because the drift detector had flagged their originating time periods; zero emails fell within the 0.30–0.70 uncertainty band, indicating that DistilBERT's near-saturated confidence profile means the routing criterion is driven almost entirely by drift flags rather than by epistemic uncertainty.

Because DistilBERT's saturated confidence profile meant that no emails fell within the 0.30–0.70 uncertainty band, a soft edge mask was applied as an alternative routing criterion for the evaluation sample. Rather than targeting mid-range uncertainty, the mask selects emails at the margins of high confidence — those with predicted spam probability in [0.85, 0.99) or ham probability in (0.01, 0.15] — capturing cases where the model is confident but not maximally so, and where LLM recalibration is most likely to be informative. Emails with probabilities outside these ranges (i.e. at the extreme ends, < 0.01 or > 0.99) are treated as sufficiently decided and bypassed. This mask was applied in conjunction with the drift-flag criterion for the 30-email sample evaluated above.

```
soft_edge_mask = (
    ((bert_test_probs >= 0.85) & (bert_test_probs < 0.99)) |  # confident spam
    ((bert_test_probs <= 0.15) & (bert_test_probs > 0.01))    # confident ham
)
```

The disagreement distribution (bottom-left figure) shows that the overwhelming majority of LLM adjustments are small (|LLM − base| < 0.10), confirming that GPT-4o-mini largely agrees with DistilBERT on clear-cut cases. Three high-disagreement cases (>0.30) drove the accuracy degradation: emails at indices 1336, 2124, and 1439 were true spam that DistilBERT correctly classified near certainty (base ≈ 0.917–0.936), but GPT-4o-mini returned low probabilities (0.10–0.20), pulling the blended final probability below the 0.5 decision threshold and causing prediction flips from correct to incorrect. No ham emails were flipped. The base

model's log-loss marginally improves under blending (0.532 → 0.527), suggesting that when the LLM does not flip predictions, it provides modest probabilistic refinement.

These results highlight a key limitation of the 60/40 weighting scheme when applied to a base model with near-saturated confidence: a small number of aggressive LLM disagreements can override correct high-confidence predictions and degrade accuracy on a small sample. A more conservative blend weight (e.g. 0.80/0.20 during confirmed drift, or conditional on disagreement magnitude) would better preserve the base model's correct high-confidence decisions while still allowing the LLM to recalibrate genuinely uncertain cases. The full-scale 400-email balanced evaluation, designed but not yet executed due to API cost constraints, will provide the statistical power needed to assess whether the log-loss improvement generalises and to tune the blend weight more rigorously.

Explainability via SHAP

Three model-appropriate explainability approaches were applied:

Model	Method	Top Signal	Generalizability	Key Limitation
Naive Bayes	Log P ratio	Mailing list tokens	Poor	No per-doc variation
Logistic Reg.	SHAP LinearExplainer	click, free, money	Moderate	Fixed token direction
DistilBERT	SHAP PartitionExplainer	Punctuation + narrative tokens	Strong	Expensive to scale

Table 9: SHAP explainability comparison across model families — method used, top discriminative signals identified, cross-domain generalisability of learned features, and key limitation of each approach.

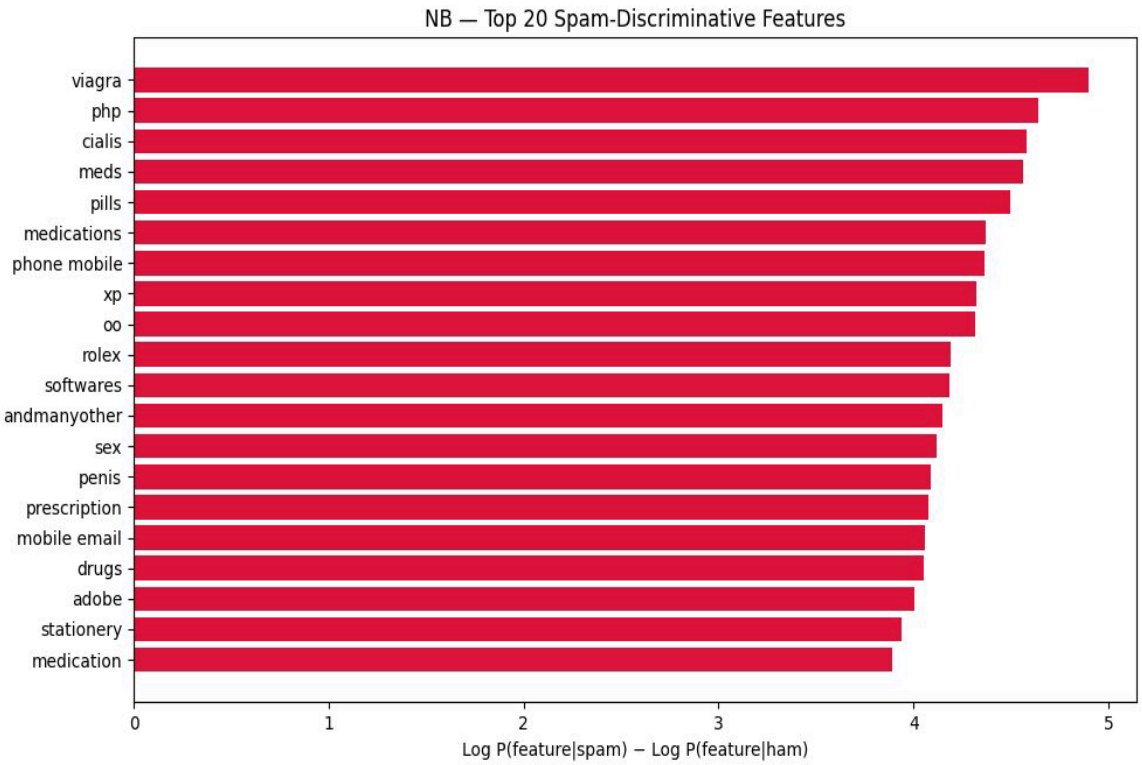

Figure 6: NB — Top 20 Spam-Discriminative Features by log P(feature|spam) − log P(feature|ham). Bars represent log-ratio; longer = more spam-indicative.

Naive Bayes explainability is computed via log-ratio analysis (log P(feature|spam) − log P(feature|ham)), which is equivalent to inspecting the model's learned class-conditional likelihood weights. The top spam-discriminative features are pharmaceutical and product terms: 'viagra' (log ratio ≈ 4.9), 'php', 'cialis', 'meds', and 'pills' all exceed 4.5. While these are semantically appropriate spam signals, the list also surfaces infrastructure artefacts: 'oo', 'xp', 'andmanyother', and 'stationery' are corpus-specific tokens that reflect Enron-era software product spam rather than generalisable spam vocabulary. The presence of 'php' as the second-ranked feature (a web scripting language term appearing in Enron-era piracy spam) further illustrates how NB learns corpus-specific lexical signatures rather than portable semantic spam indicators. This directly explains NB's collapse on the SpamAssassin OOD benchmark (accuracy 49.7%): these features do not transfer to a different email population.

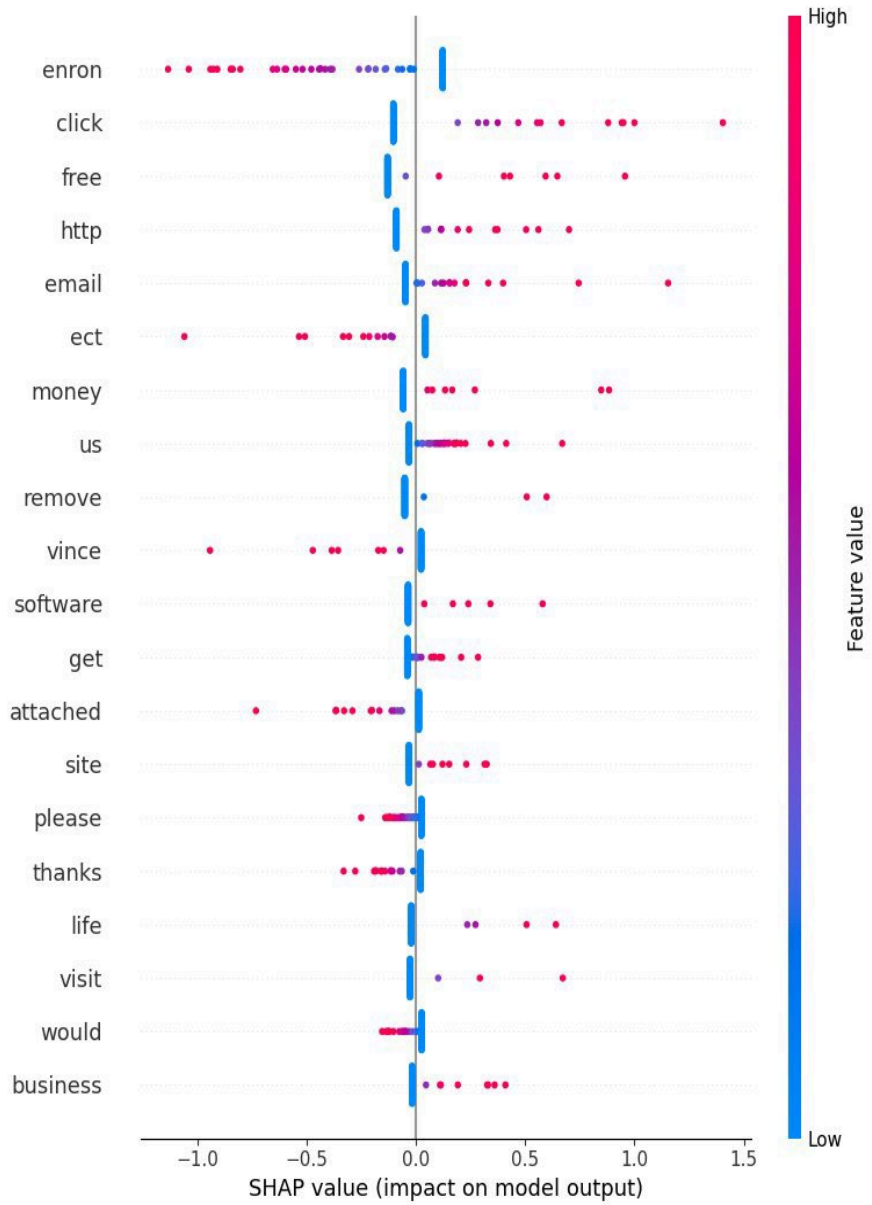

Figure 7: LR — SHAP beeswarm plot. Each dot is one test email; x-axis = SHAP value (impact on spam probability); colour = feature value (red = high, blue = low).

Logistic Regression explainability is computed via SHAP LinearExplainer, enabling per-document attribution across the test set. The beeswarm plot reveals a notably richer and more interpretable feature landscape than NB. The dominant ham signal is 'enron' (SHAP range ≈ −1.4 to +0.3): emails with high 'enron' frequency push strongly toward ham, consistent with the model learning that internal Enron communications are legitimate. 'ect' and 'vince' — Enron Energy Services and Vince Kaminski, a key Enron figure — similarly anchor the ham side, confirming that LR has learned organisation-specific ham markers. On the spam side, 'click' (SHAP up to +1.5) and 'email' (+1.3) are the strongest positive signals, followed by 'free', 'http', 'money', 'software', and 'remove'. These are semantically generalisable spam indicators that transfer more readily across email distributions — explaining LR's substantially better SpamAssassin OOD performance (84.9%) relative to NB. The wide SHAP spread for 'enron'

indicates high per-document variation, a property not visible in NB's global log-ratio analysis but important for understanding individual prediction behaviour.

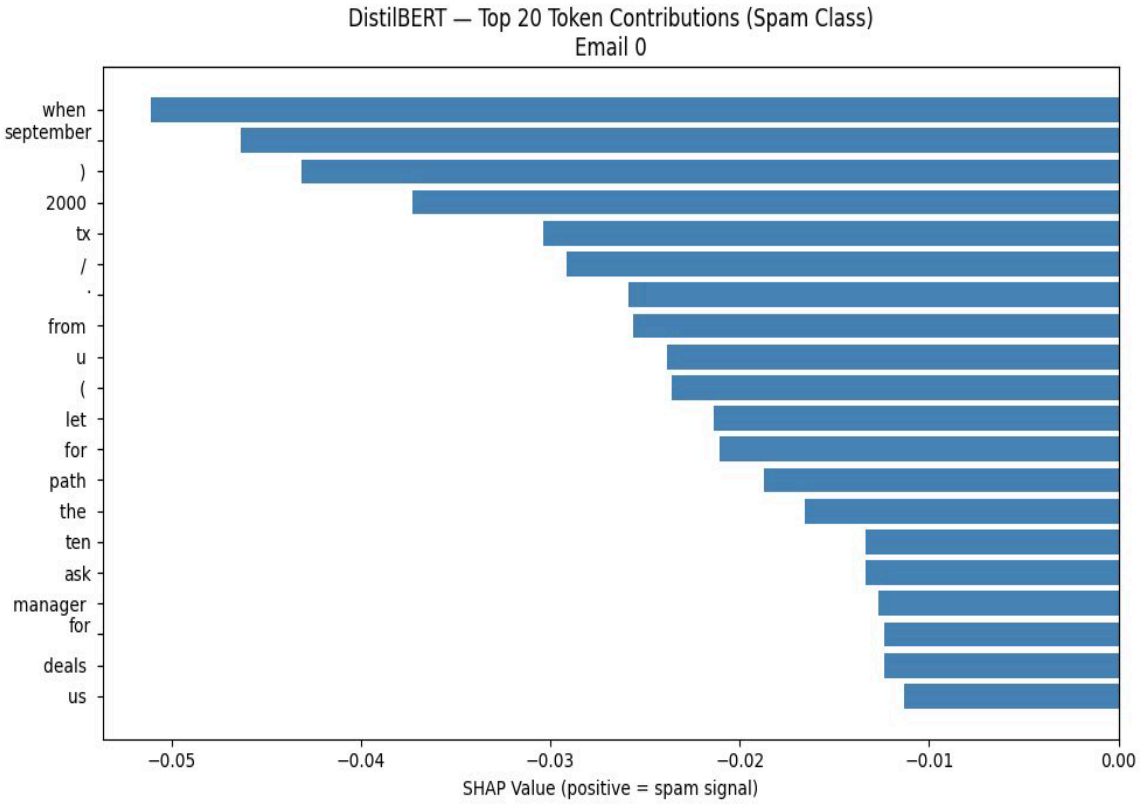

Figure 8: DistilBERT — Top 20 Token Contributions for Email 0 (Spam Class). SHAP values are negative here, indicating all listed tokens push slightly away from the spam prediction for this specific email.

DistilBERT explainability is computed via SHAP PartitionExplainer applied to individual emails, producing token-level attributions that vary per document. For Email 0, all top-20 contributing tokens have small negative SHAP values (range −0.055 to 0.00), meaning each slightly reduces the spam probability — yet the model still classified the email as spam, indicating the classification is driven by the aggregate of many small positive signals outside the top 20, or by tokens not displayed. The contextual tokens surfaced — 'when', 'september', '2000', 'tx' (Texas), 'from', 'let', 'manager', 'deals' — are coherent Enron business communication tokens that individually push slightly toward ham. This is the defining characteristic of SHAP PartitionExplainer applied to a transformer: unlike NB's global log-ratio weights or LR's linear feature attributions, DistilBERT's SHAP values are context-sensitive and email-specific. The same token can have a different SHAP value in a different email depending on surrounding context, making per-document explanations faithful to individual predictions but non-generalisable across the corpus. This per-email variability is a key strength of transformer-based explainability over bag-of-words approaches.

Discussion & Limitations

Our study demonstrates the multi-dimensional nature of robustness in spam filtering systems. There is clear evidence that, when handling adversarial perturbations and contextual language patterns, transformer models are superior; however, they remain vulnerable to distribution shifts and calibration degradation.

Model accuracy is insufficient for evaluating the reliability of security-critical machine learning systems, according to the results. It is important to take into account robustness to manipulation, stability under evolving data distributions, and the reliability of probability estimates when deploying systems in adversarial environments.

The LLM-assisted calibration layer preliminary results present a more nuanced picture than initially hypothesised. On the 30-email evaluation sample, the hybrid pipeline degraded Brier score (+0.015) and accuracy (−10 pp) relative to the base model, driven by three high-disagreement flips in which GPT-4o-mini overrode correct high-confidence DistilBERT predictions. However, log-loss marginally improved (−0.005), suggesting that probabilistic refinement is occurring on most emails and that the fixed 60/40 blend weight is too aggressive when the base model is already near-saturated. LLMs may serve effectively as a probabilistic auditing layer for uncertain predictions, but dynamic weighting conditional on disagreement magnitude is a necessary next step before the hybrid pipeline can be recommended for deployment.

Limitations

After interpreting the results, there are several limitations that should be considered. The first limitation concerns cross-corpus evaluation design. Because the Enron corpus is structured such that ham clusters in 2000–2002 and spam in 2003–2005, a temporal split is not viable; the stratified split used instead controls class composition within the Enron evaluation but does not resolve the class distribution confound in external OOD benchmarks such as the Bruce 2021 spam-only corpus. Some observed performance degradation on external corpora reflects class composition differences alongside domain shift, and cannot be attributed purely to semantic evolution in spam content.

A second limitation is that the external Bruce spam dataset only contained spam emails. This limited the ability of the model to evaluate both classes that are needed for some calibration metrics. The Brier score shifts provide useful information on reliability degradation, as a fully balanced evaluation would allow for a more precise calibration analysis.

A final limitation that should be highlighted is that the hybrid LLM calibration experiment was conducted on a limited evaluation sample. The results from this experiment should be interpreted as preliminary evidence instead of definitive conclusions.

Supplementary Research

Concept Drift Detection

The study introduces an unsupervised drift monitoring framework to detect distribution shifts without requiring labeled data.

Monthly batches are created from incoming emails. A 15-topic Latent Dirichlet Allocation model, trained on the Enron training corpus, maps each email to a topic distribution vector. The mean topic distribution per monthly batch is then compared against the baseline training distribution.

Jensen-Shannon divergence — a symmetric, bounded measure of difference between probability distributions — is computed between each batch distribution and the training baseline. When JS divergence exceeds a threshold of 0.25, the batch is flagged as a potential drift period. This threshold was selected empirically to balance sensitivity to genuine topical shifts against false positives from within-domain variation.

This approach allows the system to identify semantic changes in incoming email streams before classification accuracy visibly degrades. Across 35 monthly batches evaluated (2000-03 to 2005-07), 18 were flagged as drifted (51.4%), with peak JS divergence of 0.409 in December 2001, coinciding with the Enron corporate crisis.

RQ2: Semantic Embeddings & Concept Drift Detection Results

Drift Detection Pipeline

Three-stage unsupervised drift detection was implemented: (1) Sentence-BERT (all-MiniLM-L6-v2) encodes each email into a 384-dimensional semantic vector; (2) LDA models the training corpus as a 15-topic distribution; (3) Jensen-Shannon divergence monitors monthly batches of incoming email against the training baseline, flagging drift when JS divergence exceeds 0.25.

Results

Across 35 monthly batches spanning the full Enron email archive (2000–2002 and 2004–2005), 18 batches (51.4%) exceeded the drift threshold of JS > 0.25, with 17 stable batches below it. The maximum divergence of 0.409 occurred in December 2001, coinciding with the most acute phase of Enron's internal crisis and a corresponding surge in operational email traffic (Topic 4: scheduling, hourahead, California) that diverged sharply from the mixed training baseline. January 2002 (JS = 0.399) represents the tail of the same event. Notably, the system detected drift in all nine months of 2000 evaluated (March–December), driven by Topic 14 (ect, enron, 2000, hou, vince) — early Enron internal communications that differ topically from the training distribution despite belonging to the same organisation. This indicates the LDA-based

detector is sensitive to topical composition shifts even within-domain, a property that could trigger false alarms in deployment if the threshold is not calibrated against expected within-distribution variance.

Figure 9: Three-panel semantic drift monitor. Panel 1: Jensen-Shannon divergence per monthly batch vs. 0.25 threshold (red dashed); shaded = drift-flagged periods. Panel 2: Mean classifier confidence (NB orange, LR green, DistilBERT purple) across drift periods, with 0.5 decision threshold (grey dotted). Panel 3: Batch spam rate (blue = stable, red = drift-flagged).

Panel 2 of the figure surfaces a key diagnostic not visible from accuracy metrics alone: classifier confidence behaviour across drift periods. NB (orange) and LR (green) maintain mean confidence near 1.0 throughout the 2000–2002 batches, even during the nine drift-flagged periods in 2000 and the Enron crisis spike in 2001-12 (JS = 0.409). This falsely stable confidence is precisely the failure mode that makes concept drift dangerous in deployment — the model appears certain while predicting on a shifted distribution. In contrast, DistilBERT (purple) shows near-zero mean spam probability across all 2000–2001 batches (reflecting that these are

predominantly ham-period emails), then jumps sharply to near-1.0 from 2002-01 onward as the corpus transitions to all-spam batches. LR shows a more pronounced confidence drop in the 2004–2005 OOD period (falling to ~0.75–0.80), consistent with its broader uncertainty response to distribution shift documented in the calibration analysis. Panel 3 confirms the ham/spam asymmetry: 2000–2001 batches are near-zero spam rate, meaning early drift flags are within-domain topical variation rather than spammer adaptation, while the 2004–2005 flagged batches (red bars) coincide with genuine spam evolution in an all-spam period — validating the drift detector's sensitivity to topical composition change across both class regimes.

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